
Multiplierless 2D Input Transformation Unit (input_transform_2d.sv)
Input Tile d4 × 4 INT16 (Signed)
16 × 16b
4 × 1D Column Transforms input_transform_1d $W = B^T \cdot d$ $w_0 = d_0 - d_2$ $w_1 = d_1 + d_2$ $w_2 = d_2 - d_1$ $w_3 = d_1 - d_3$ Bit-growth: 16 → 17 bits
16 × 17b
Spatial TransposeRewiring $W^T = (B^T d)^T$ Parallel bus permutation(Zero logic overhead)
16 × 17b
4 × 1D Row Transforms input_transform_1d $V = B^T \cdot W^T$ $v_0 = w_0 - w_2$ $v_1 = w_1 + w_2$ $v_2 = w_2 - w_1$ $v_3 = w_1 - w_3$ Bit-growth: 17 → 18 bits
16 × 18b
PIPE REG (18b)
16 × 18b
Transformed Tile V4 × 4 INT18 (Signed)